

Experimental Design

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Experimental Design

Hypothesis Errors, Power and Independence

👉 Errors in Hypothesis Testing.

- ◇ Type I error and the false positive.
- ◇ Type II error and the false negative.
- ◇ The Neyman Pearson Lemma
- ◇ Power and the likelihood of detecting an effect.

👉 Independent measurements and DOE.

👉 Experimental Controls.

Type I Error: Type I Error (False Positive)

- ◇ **Type I Error** occurs when we reject the null hypothesis (H) when it is actually **true**.

It is also called a **false positive**.

Example: A healthy individual tests positive for a disease.

- ◇ **Probability of Type I Error:** α

α is typically set at 0.05 (or 5%), indicating a 5% probability of committing a **Type I error**.

Type I Error (False Positive)

- ◇ A **Type I Error** occurs when we **reject the null hypothesis (H_0)** even though it is **actually true**.

It is also known as a **false positive**.

◇ Definition

$$\text{Type I Error} = P(\text{Reject } H_0 \mid H_0 \text{ is true})$$

◇ Example

◇ Medical Testing Example:

A person who is **actually healthy** is diagnosed as **having the disease** based on the test result.

- True condition: Healthy
- Test decision: Disease detected

➡ This is a **Type I Error (false positive)**



◆ Probability of Type I Error

The probability of committing a Type I Error is denoted by α .

$$\alpha = P(\text{Type I Error})$$

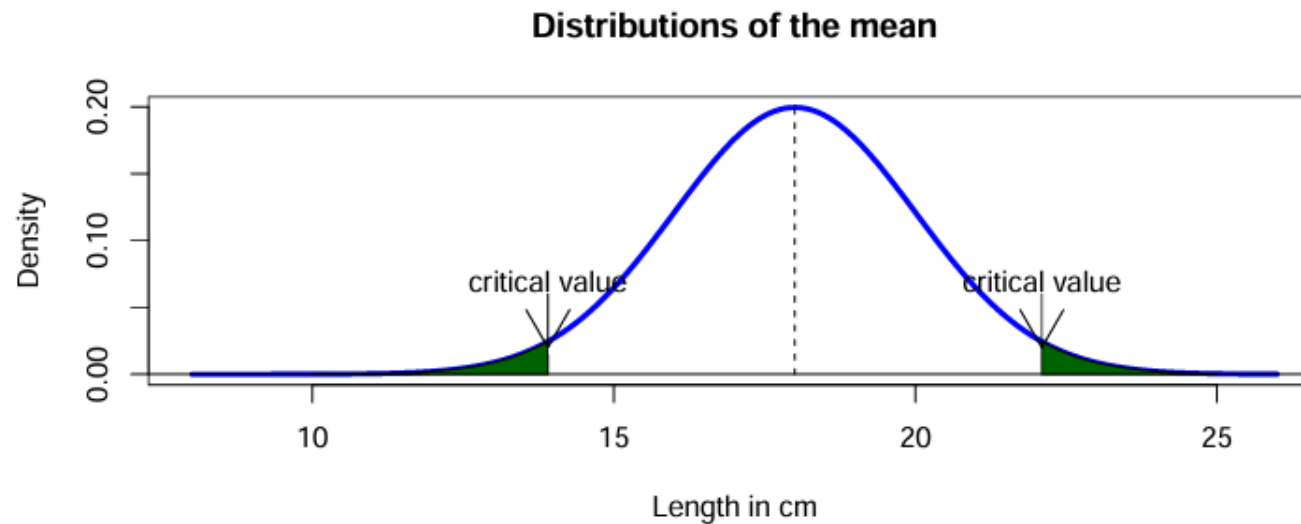
- Commonly chosen values: **0.05, 0.01, 0.10**
- $\alpha = 0.05$ means: There is a **5% chance of rejecting a true null hypothesis**

◆ Interpretation of $\alpha = 0.05$

If the same hypothesis test is repeated many times under identical conditions:

- About **5 out of 100 tests** will incorrectly reject H_0
- Even when H_0 is true

Type I Error (intuition)



Challenge Type I: Multiple Testing

- ◇ α can be set by researcher.
- ◇ As the number of tests increases, the probability of getting at least one false positive increases.
- ◇ **Family-wise Error Rate (FWER):** The probability of making at least one Type I Error among m tests.

$$FWER = 1 - (1 - \alpha)^m$$

 **Family-Wise Error Rate (FWER)** is the probability of making **at least one Type I error (false positive)** when performing **multiple hypothesis tests**.

Definition

If you conduct **m** independent hypothesis tests, each at significance level **α** , then:

$$FWER = P(\text{at least one Type I error among } m \text{ tests})$$

Formula: Assuming the tests are **independent**:

$$FWER = 1 - (1 - \alpha)^m$$

- $(1 - \alpha)$ = probability of **not** making a Type I error in one test
- $(1 - \alpha)^m$ = probability of **no Type I errors in all m tests**
- $1 - (1 - \alpha)^m$ = probability of **at least one Type I error**

Example : Family-Wise Error Rate (FWER)

If $\alpha = 0.05$ and $m = 10$ then $FWER$

$$FWER = 1 - (1 - \alpha)^m = 1 - (1 - 0.05)^{10} = 1 - (0.95)^{10} \approx 0.401$$

So, there is about a **40.1% chance** of making **at least one false rejection** across the 10 tests.

 As the number of tests m increases, **FWER increases**, which is why multiple testing corrections (like **Bonferroni correction**) are used to control FWER.

Some Example : Family-Wise Error Rate (FWER):

The **Family-Wise Error Rate (FWER)** is the probability of making **at least one Type I error** when performing m hypothesis tests at significance level α

$$FWER = 1 - (1 - \alpha)^m$$

1. Simple Linear Regression

(2 degrees of freedom $\rightarrow$ 2 hypothesis tests)

Number of tests: $m = 2$ and Significance level: $\alpha = 0.05$

$$FWER = 1 - (1 - \alpha)^m = 1 - (1 - 0.05)^2 = 1 - (0.95)^2 \approx 0.098$$

Interpretation: There is about a **9.8% chance** of committing at least one Type I error.

2. One-Way ANOVA

(One factor with 5 levels $\rightarrow$ 5 hypothesis tests)

Number of tests: $m = 5$ and Significance level: $\alpha = 0.05$

$$FWER = 1 - (1 - \alpha)^m = 1 - (1 - 0.05)^5 = 1 - (0.95)^5 \approx 0.226$$

Interpretation: There is about a **22.6% chance** of at least one false positive.



3. Two-Way ANOVA with Interaction

(Factor A: 3 levels, Factor B: 4 levels $\rightarrow$ 20 hypothesis tests)

Number of tests: $m = 20$ and Significance level: $\alpha = 0.05$

$$FWER = 1 - (1 - \alpha)^m = 1 - (1 - 0.05)^{20} = 1 - (0.95)^{20} \approx 0.64$$

Interpretation: There is about a **64% chance** of making at least one Type I error.

4. High-Throughput Experiments

(Microarrays, mass spectrometry, genomics)

Number of tests: $m = 150$ and Significance level: $\alpha = 0.05$

$$FWER = 1 - (1 - \alpha)^m = 1 - (1 - 0.05)^{150} = 1 - (0.95)^{150} \approx 0.999$$

Interpretation: There is an **almost certain probability (99.9%)** of at least one false positive.

Or stated otherwise:

- ◇ There is a 99.9% chance of having at least one false positive in 150 tests with $\alpha = 0.05$
- ◇ There is a 77.9% chance of having at least one false positive in 150 tests with $\alpha = 0.01$
- ◇ There is a 13.9% chance of having at least one false positive in 150 tests with $\alpha = 0.001$

Interpretation of FWER for Multiple Testing ($m = 150$)

When performing **150 independent hypothesis tests**, the **Family-Wise Error Rate (FWER)** represents the probability of obtaining **at least one false positive**.

$$FWER = 1 - (1 - \alpha)^{150}$$

◇ Significance level $\alpha = 0.05$

$$FWER = 1 - (1 - 0.05)^{150} = 1 - (0.95)^{150} \approx 0.999$$

Statement:

There is a **99.9% chance** of obtaining **at least one false positive** among 150 tests when each test is conducted at the 5% significance level.

◇ Significance level $\alpha = 0.01$

$$FWER = 1 - (1 - 0.01)^{150} = 1 - (0.99)^{1500} \approx 0.779$$

Statement:

There is a **77.9% chance** of obtaining **at least one false positive** among 150 tests when each test is conducted at the 1% significance level.

◇ Significance level $\alpha = 0.001$

$$FWER = 1 - (1 - 0.001)^{150} = 1 - (0.999)^{1500} \approx 0.139$$

Statement:

There is a **13.9% chance** of obtaining **at least one false positive** among 150 tests when each test is conducted at the 0.1% significance level.

 Even very small per-test significance levels can lead to a **substantial probability of false discoveries** when the number of tests is large. This is why **FWER control** (e.g., Bonferroni correction) or **FDR control** methods are essential in high-throughput data analysis such as genomics and proteomics.

Bonferroni Correction

When multiple hypothesis tests are performed, the **Family-Wise Error Rate (FWER)** increases.

The **Bonferroni correction** is a simple and conservative method to **control FWER**.

Basic Idea

If k hypothesis tests are conducted, adjust for multiple testing by either:

1. Adjusting the p-values

Reject H_0 if :

$$\mathcal{P}_{adjusted} = \mathcal{P}_{original} \times k$$

$$\mathcal{P}_{adjusted} \leq \alpha$$

2. Adjusting the significance level

Reject H_0 if :

$$\alpha_{Bonferroni} = \frac{\alpha}{k}$$

$$\mathcal{P}_{original} \leq \frac{\alpha}{k}$$

Control of FWER

Under Bonferroni correction, the FWER satisfies:

$$FWER \leq k \times \frac{\alpha}{k} = \alpha$$

Hence, the probability of making **at least one Type I error** is controlled at the desired level α .

Example 1: Overall Error Control

Suppose:

Number of tests: $k = 10$

Desired FWER: $\alpha = 0.05$

Bonferroni-adjusted significance level:

$$\alpha_{Bonferroni} = \frac{\alpha}{k} = \frac{0.05}{10} = 0.005$$

Approximate FWER:

$$FWER = 1 - (1 - \alpha_{Bonferroni})^k = 1 - (1 - 0.005)^{10} \approx 0.05$$

Interpretation: The Bonferroni correction keeps the overall Type I error rate close to **5%**.



Example 2: Adjusted p-value ($\mathcal{P}_{adjusted}$)

Suppose:

Original p-value: $p = 0.00008$

Number of tests: $k = 1000$

Bonferroni-adjusted p-value:

$$\mathcal{P}_{adjusted} = \mathcal{P}_{original} \times k = 0.00008 \times 1000 = 0.08$$

Conclusion:

Since $\mathcal{P}_{adjusted} = 0.08 > 0.05$, the null hypothesis is **not rejected**, even though the original p-value appeared highly significant.



Impact on Type II Error

-  Bonferroni correction is **very conservative**
-  Fewer null hypotheses are rejected
-  **Type I errors decrease**
-  **Type II errors increase** (loss of power)

This trade-off becomes severe when k is large.

Interpretation of the R Code

```
➤ k <- 4:50
➤ plot(1 - (1 - 0.05/k)^k ~ k,
      xlab = "k(number of tests)",
      ylab = "Overall value for alpha")
```



Errors in Hypothesis Testing: Type I Error



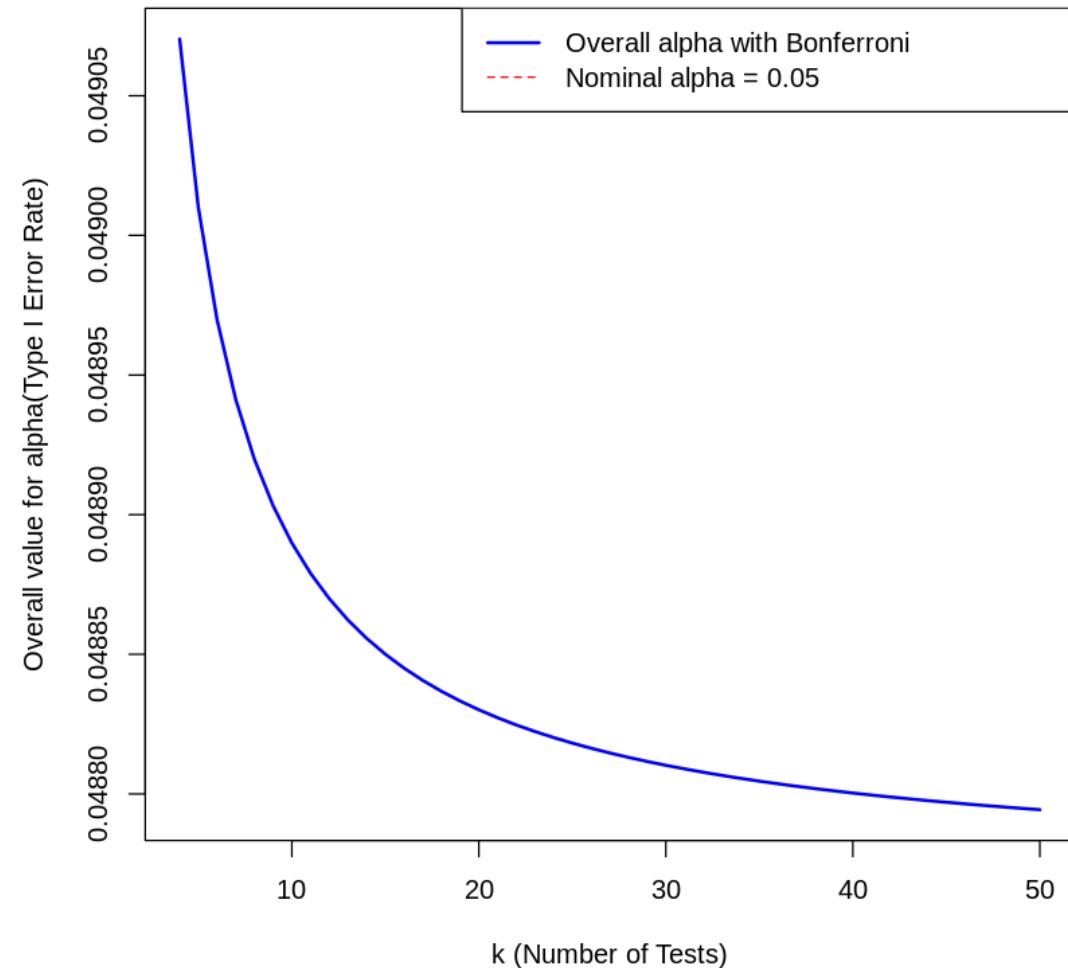
Explanation

- For each value of k , the per-test significance level is adjusted to $\frac{0.05}{k}$
- The expression

$$1 - \left(1 - \frac{0.05}{k}\right)^k$$

- computes the **overall Type I error rate**
- The plot shows that as k increases, the **overall α remains close to 0.05**

Bonferroni Correction Effect



`influenceIndexPlot` in Linear Regression

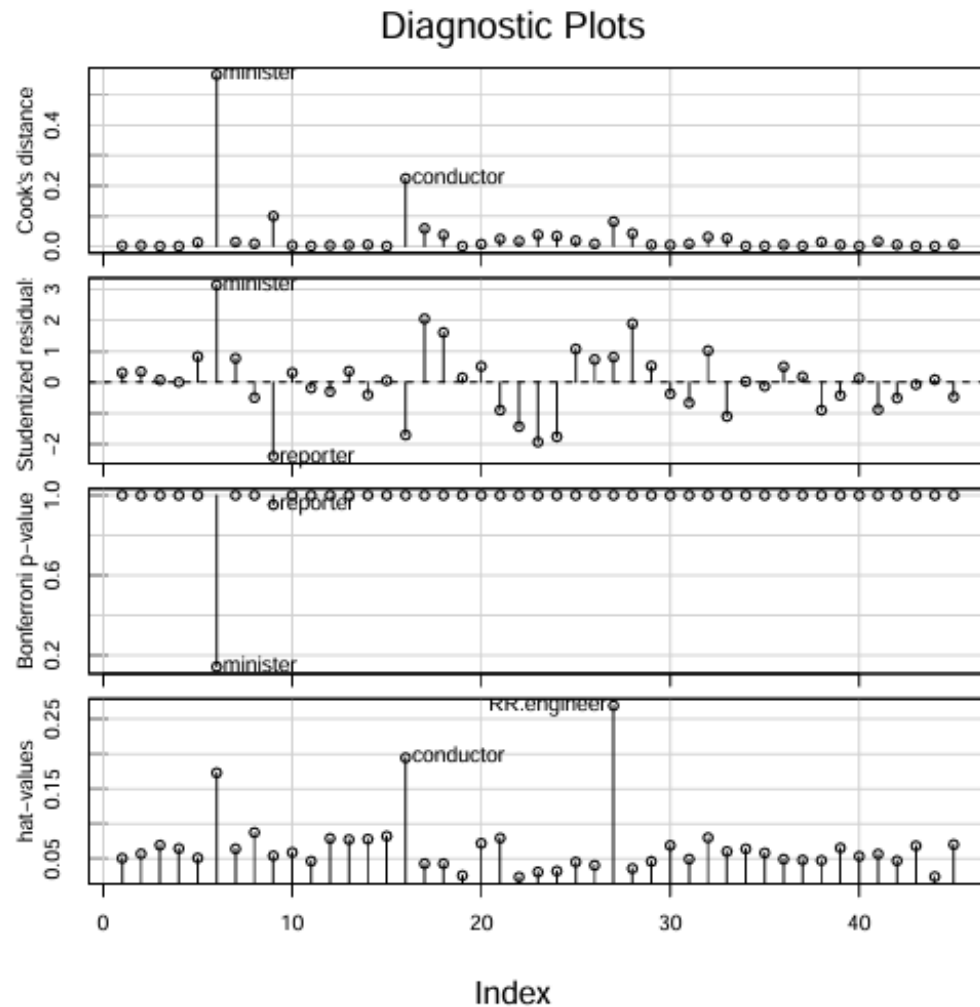
R Code

- `library(car)`
- `Model <- lm(prestige ~ income + education, data = Duncan)`
- `par(mar = c(3.1, 4.1, 1.1, 2.1))`
- `influenceIndexPlot(Model)`



InfluenceIndexPlot Shows

The plot displays **four panels**, each indexed by observation number:



1. Cook's Distance
2. Studentized Residuals
3. Bonferroni p-value
4. Hat Values (Leverage)

Interpretation

- **High Cook's Distance** → influential observation
- **Large Studentized Residual + small Bonferroni p-value** → response outlier
- **High Leverage** → unusual predictor values
- Observations large in **multiple panels** deserve special attention

Example: `outlierTest()` in Linear Regression

➤ `outlierTest(Model)`

R Output:

```
No Studentized residuals with Bonferroni p < 0.05
Largest |rstudent|:
      rstudent unadjusted p-value Bonferroni p
minister 3.134519      0.0031772      0.14297
```

This statement means:

No observation is a statistically significant outlier after applying the Bonferroni correction at the 5% level.

Quantity	Value
Largest	$(r_{student}) = 3.134519$
Unadjusted p -value	0.0031772
Bonferroni p -value	0.14297



1. Studentized Residual ($r_{student}$)

- Measures how far an observation's response value deviates from the fitted model.
- A value of **3.13** is **large**, suggesting a **potential outlier** if judged alone.

2. Unadjusted p -value

$$p = 0.0031772$$

- This p -value tests:

H_o : The observation is not an outlier

- Without correction, this observation **would be considered significant** at $\alpha=0.05$.

3. Bonferroni Correction

Since **multiple observations are tested simultaneously**, Bonferroni correction is applied:

$$\mathcal{P}_{Bonferroni} = p \times n$$

Here:

- $p = 0.0031772$
- Number of observations $n = 45$

$$\mathcal{P}_{Bonferroni} = 0.0031772 \times 45 = 0.14297$$



4. Bonferroni p-value Interpretation

- $\mathcal{P}_{Bonferroni} = 0.14297 > 0.05$
- Hence, the observation is **not statistically significant** after correction.
- This explains the message:

No Studentized residuals with Bonferroni $p < 0.05$

👉 Bonferroni correction provides strong protection against false positives but may fail to detect real outliers due to loss of power.

False Discovery Rate (FDR)

The **False Discovery Rate (FDR)** is the **expected proportion of false positives among all rejected null hypotheses**.

Formal Definition

$$\text{FDR} = E \left(\frac{\text{Number of false rejections}}{\text{Total number of rejections}} \right)$$

This is fundamentally different from FWER, which controls the probability of **any** false positive.

Logic Behind FDR

The logic behind the **False Discovery Rate** is as follows::

- If **many variables are significant**, then surely there must be **some true effects**
- The actual chance of a **false positive** is then much lower than the **FWER** suggests

Possible procedure for FDR

- Rank all **p-values** from **low to high**
- Multiply each **p-value** by the fraction $\frac{k}{i}$
- Where k is the number of tests and i the rank of the p -value

$$\mathcal{P}_{FDR_i} = p_i \times \frac{k}{i}$$

- There are many more multiple testing correction methods.



Benjamini–Hochberg (BH) Procedure for FDR

A commonly used FDR procedure is the **Benjamini–Hochberg method**.

Step-by-Step Procedure

1. Perform k hypothesis tests and obtain p-values

$$p_1, p_2, \dots, p_k$$

2. Rank the p-values from smallest to largest:

$$p_1 \leq p_2 \leq \dots \leq p_k$$

3. Compute the **FDR-adjusted p-values**:

$$\mathcal{P}_{FDR_i} = p_i \times \frac{k}{i}$$

where:

k = total number of tests

i = rank of the p-value

4. Compare $\mathcal{P}_{FDR_i}$ with α

5. Reject all hypotheses with: $\mathcal{P}_{FDR_i} \leq \alpha$



Interpretation of FDR

- If FDR is controlled at $\alpha = 0.05$

Among all rejected hypotheses, about 5% are expected to be false positives.

- This is a **much more realistic criterion** when thousands of tests are performed.

False Discovery Rate vs FWER

Criterion	FWER	FDR
Controls	Probability of ≥ 1 false positive	Proportion of false positives
Strictness	Very conservative	Less conservative
Power	Low	Higher
Typical use	Clinical trials, confirmatory studies	Genomics, proteomics, ML



Example

- 1000 genes tested
- 200 genes declared significant
- $FDR = 0.05$



On average, about **10 of the 200 significant genes** are expected to be false discoveries.



Important Points



FDR controls the **rate of false discoveries**, not their absolute occurrence



More powerful than Bonferroni



Widely used in **high-dimensional data analysis**



Especially suitable when **some false positives are acceptable**



False Discovery Rate (FDR) is the expected proportion of false positives among rejected hypotheses and is commonly controlled using the **Benjamini–Hochberg** procedure.



Type II error: Type II Error (False Negative)

- **Type II Error** occurs when we fail to reject the null hypothesis (H) when it is actually **false**.
- It is also called a **false negative**.
 - **Example:** A medical test shows a person is healthy when they actually have the disease.
- **Probability of Type II error:** β
- Power of a test is defined as $1 - \beta$, which is the probability of correctly rejecting a false null hypothesis.

Type II Error (False Negative)

- ◇ A **Type II Error** occurs when we **fail to reject the null hypothesis (H_0)** even though it is **actually false**.

It is also known as a **false negative**.

- ◇ **Definition**

$$\text{Type II Error} = P(\text{Fail to reject } H_0 \mid H_0 \text{ is false})$$

- ◇ **Example**

- ◇ **Medical Testing Example:**

A person who **actually has a disease** is diagnosed as **healthy** by the test.

- True condition: Diseased
- Test decision: No disease detected

➡ This is a **Type II Error (false negative)**



◇ Probability of Type II Error

The probability of committing a Type II Error is denoted by β .

$$\beta = P(\text{Type II Error})$$

◇ Power of a Test

The **power of a statistical test** is defined as:

$$\text{Power} = 1 - \beta$$

- Power is the probability of **correctly rejecting a false null hypothesis**
- A **high-power test** is more likely to detect a real effect

◇ Interpretation

- Large $\beta \rightarrow$ high chance of missing a true effect
- Small $\beta \rightarrow$ high test power

Commonly desired power:

$$\text{Power} \geq 0.80$$

This means there is at least an **80% chance** of detecting a true effect.



◆ Relationship with Type I Error

- α controls **false positives**
- β controls **false negatives**
- Reducing α often **increases** β unless sample size is increased

👉 **Type II Error** is the failure to reject a false null hypothesis; its probability is β , and the ability to avoid this error is measured by test **power** $1 - \beta$



Conditional Probabilities in Hypothesis Testing

In hypothesis testing, errors are defined using **conditional probabilities** — that is, probabilities conditioned on whether the null hypothesis H_0 is true or false.

Type I Error	$P(\text{Reject } H_0 \mid H_0 \text{ is TRUE}) = \alpha$
Type II Error	$P(\text{Fail to reject } H_0 \mid H_0 \text{ is FALSE}) = \beta$

Power of the Test

$$\text{Power} = 1 - \beta$$

$$\text{Power} = P(\text{Reject } H_0 \mid H_0 \text{ is FALSE}) = 1 - \beta$$

Power is the probability of correctly detecting a true effect.

Decisions H_0 testing

Possible outcomes in hypothesis testing

Reality	Decision	Result	Probability
H_0 True	Reject	Type I Error	α
H_0 False	Fail to Reject	Type II Error	β
H_0 False	Reject	Correct Decision	$(1 - \beta)$ (Power)
H_0 True	Fail to Reject	Correct Decision	$(1 - \alpha)$



Errors in Hypothesis Testing: Type II Error

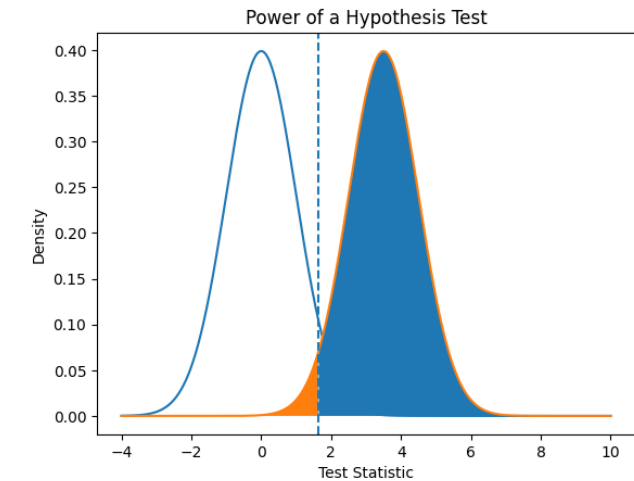
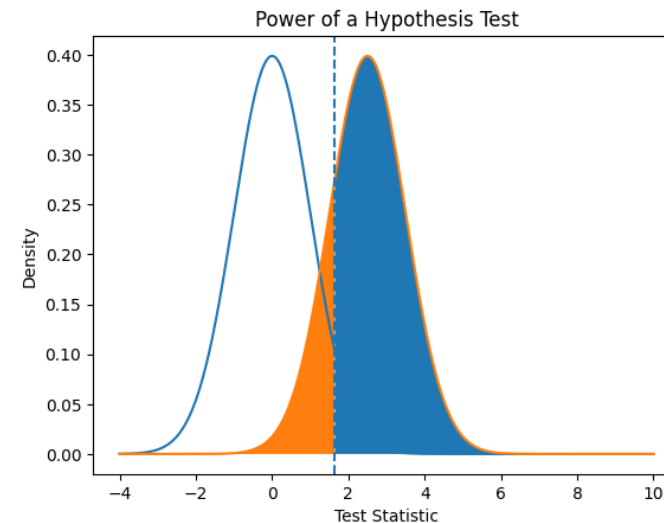
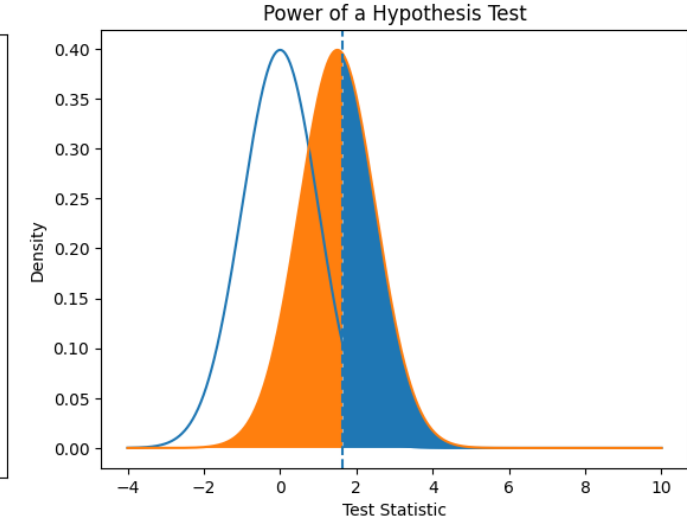
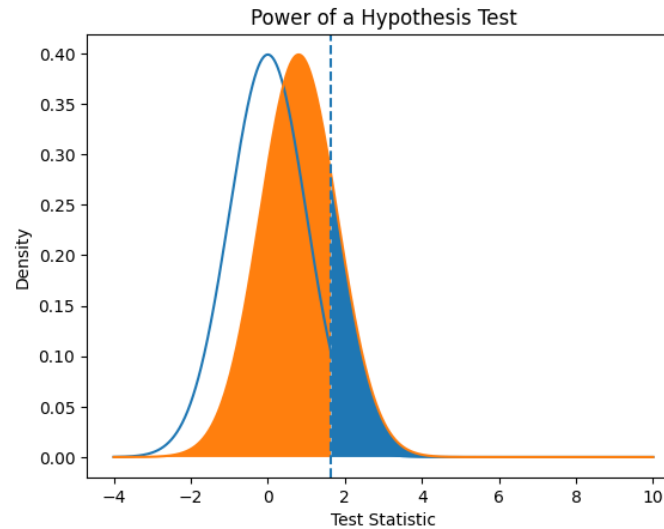


◆ The four **Power of a Hypothesis Test** diagrams showing:

- Null distribution
- Alternative distribution
- Critical value ($\alpha = 0.05$)
- Type II error region (β)
- Power region ($1 - \beta$)

As the separation between the **null** and **alternative** means **increases**, you can clearly see:

- Type II error **decreases**
- Power **increases**





Hypothesis Testing Decision Table

Decision Based on Data	H_0 is True	H_0 is False
Reject H_0	Type I Error (False Positive) α	Correct Rejection (Power) $(1 - \beta)$
Do not Reject H_0	Correct (True Negative) $(1 - \alpha)$	Type II Error (False Negative) β

◆ Interpretation

- **Type I Error (α)** → Rejecting a true H_0
- **Type II Error (β)** → Failing to reject a false H_0
- **Power ($1 - \beta$)** → Correctly rejecting a false H_0
- **True Negative ($1 - \alpha$)** → Correctly not rejecting a true H_0

Neyman-Pearson Lemma:

A framework for hypothesis testing by maximizing the **power of a test** while controlling for **Type I error**.

◇ Hypothesis Testing Framework :

We consider **simple hypotheses**:

- **Null Hypothesis (H_0)**: The assumption that there is no effect. **Or** ($H_0: \theta = \theta_0$)
- **Alternative Hypothesis (H_1)**: The assumption that there is an effect **Or** ($H_1: \theta = \theta_1$)

where both hypotheses specify a single distribution.

◇ Key Result of the Neyman–Pearson Lemma:

The **most powerful test** for a **given significance level α** (probability of Type I error) is the one that maximizes the probability of rejecting H_0 when H_0 is true.

Maximize $P(\text{Reject } H_0 \mid H_0 \text{ is FALSE})$

Likelihood Ratio Test (LRT)

Define the likelihood functions:

$$L(H_0) = f(x | H_0)$$

$$L(H_1) = f(x | H_1)$$

The likelihood ratio is:

$$\Lambda(x) = \frac{L(H_0)}{L(H_1)} = \frac{f(x | H_0)}{f(x | H_1)}$$

The Neyman–Pearson test rejects H_0 when:

$$\frac{f(x | H_0)}{f(x | H_1)} > k$$

for some constant k chosen such that:

$$P(\text{Reject } H_0 | H_0 \text{ is TRUE}) = \alpha$$

What Is Being Maximized?

The lemma ensures that we maximize:

$$P(\text{Reject } H_0 \mid H_0 \text{ is TRUE}) = \alpha$$

This is the **power** of the test.

So the correct maximization statement is:

$$\text{Maximize } P(H_0 \text{ Rejected} \mid H_1 \text{ TRUE})$$

 **Not:** $P(H_0 \text{ Rejected} \mid H_0 \text{ FALSE})$

Neyman–Pearson Lemma and the Likelihood Ratio Test (LRT)

◇ Connection Between NP Lemma and LRT

The **Neyman–Pearson Lemma** states that:

For testing a **simple null hypothesis** against a **simple alternative hypothesis**, the **Likelihood Ratio Test (LRT)** is the most powerful test at a given significance level α .

Thus:

- For simple hypotheses → **LRT is optimal**
- It maximizes **power** while controlling **Type I error**

◆ Likelihood Ratio

The **likelihood ratio** is defined as:

$$\Lambda = \frac{L(H_0)}{L(H_1)}$$

or equivalently,

$$\Lambda = \frac{P(\text{Data} \mid H_0)}{P(\text{Data} \mid H_1)}$$

where:

$L(H_0)$ = likelihood of data under the null hypothesis

$L(H_1)$ = likelihood of data under the alternative hypothesis

◆ Decision Rule of LRT

The test rejects H_0 when:

$$\Lambda(x) \leq c$$

for some constant c chosen so that:

$$P(\text{Reject } H_0 \mid H_0 \text{ is TRUE}) = \alpha$$

◆ Interpretation:

- If the ratio Λ is **large** $\rightarrow$
 $L(H_0) \gg L(H_1) \rightarrow$ Data supports $H_0 \rightarrow$ **Do not reject**
- If the ratio Λ is **small** $\rightarrow$
 $L(H_1) \gg L(H_0) \rightarrow$ Data supports $H_1 \rightarrow$ **Reject H_0**

So the rejection region corresponds to data that is **much more likely under H_1** .

- ◆ **Control of Type I Error:** Type I Error is controlled by the significance level of the test (α usually set at 0.05)

The significance level α controls: $P(\text{Reject } H_0 \mid H_0 \text{ TRUE}) = \alpha = 0.05$

This ensures the probability of a **false positive** is controlled.

- ◆ **Maximizing Power:** The Neyman-Pearson Lemma ensures that, for a given α , the test has the maximum power (i.e., the lowest Type II Error β).

The Neyman–Pearson Lemma guarantees that:

$$\text{Power} = P(\text{Reject } H_0 \mid H_1 \text{ TRUE}) = 1 - \beta$$

is **maximized** among all tests that satisfy:

$$P(\text{Reject } H_0 \mid H_0 \text{ TRUE}) = \alpha$$

Thus:

- ✓ Type I error is fixed
- ✓ Type II error β is minimized
- ✓ Power $(1 - \beta)$ is maximized



What is Power Analysis?

Power analysis is a statistical method used to determine:

- The probability that a test will correctly reject a false null hypothesis.
- The required **sample size** to detect a meaningful effect.

Mathematically,

$$\text{Power} = P(\text{Reject } H_0 \mid H_1 \text{ TRUE}) = 1 - \beta$$

where:

- α = Type I error probability
- β = Type II error probability
- $1 - \beta$ = Power



Why power analysis?

- Power analysis is a crucial step in research design.
- Helps to determine an adequate sample size.
- Ensures that studies are well-designed and capable of detecting meaningful effects.
- Hence, enhancing the reliability of research.

Types of Power Analysis

- **A Priori Power Analysis:** Conducted before data collection to estimate the required sample size.
- **Post Hoc Power Analysis:** Conducted after a study to assess its power based on the obtained effect size
- **Sensitivity Analysis:** Examines how different effect sizes impact the power of a study.



A Priori Power Analysis

- Fix probability for **type 1 error** with α : e.g. 0.05
- Decide on **minimal power**: e.g 0.8
- What is the minimal biological significant effect size.
- Estimate the **expected residual variance of SD**
 - literature.
 - similar experiments.
 - pilot study.
- For a **priori power analysis** calculate **sample size (n)** to obtain **minimal power**.
- For a **post-hoc power analysis** n is known and the **power** is calculate.



Steps in A Priori Power Analysis

To perform A Priori power analysis:

Step 1: Fix Type I Error

$\alpha = 0.05$ (commonly used)

Step 2: Decide Desired Power

Usually:

Power = 0.80 or 0.90

Power = 0.80 means: 80% chance of detecting a real effect.

Step 3: Specify Minimum Meaningful Effect Size

This is the **practically significant effect**, not just statistically significant.

Example:

- Mean difference
- Correlation coefficient
- Odds ratio

Effect size can be obtained from:

- Literature
- Previous studies
- Similar experiments
- Pilot study



Step 4: Estimate Variance (or Standard Deviation)

Needed because:

Standard error depends on variance

Variance estimate comes from:

- Literature
- Similar experiments
- Pilot study

Step 5: Calculate Required Sample Size (n)

Using statistical formulas or software (R)

For example (one-sample mean test):

$$n = \left(\frac{Z_{\alpha} + Z_{\beta}}{\text{Effect Size}} \right)^2 \sigma^2$$

This gives the minimum n needed to achieve desired power.



Post-Hoc Power Calculation using `power.t.test()` in R

What is Post-Hoc Power Analysis?

Post-hoc power analysis is performed **after the study is completed**.

we already:

- Collected data
- Performed a t-test
- Obtained an effect size

To calculate the **power** to see: “Was my sample size large enough to reliably detect the effect?”

 **R Code:**

➤ `power.t.test(n = 15, delta = 1, sig.level = 0.05, alternative = "two.sided")`

Output Interpretation

Two-sample t test power calculation

n = 15

delta = 1

sd = 1

sig.level = 0.05

power = 0.752921

alternative = two.sided

NOTE: n is number in **each** group



Each Parameter Means:

- ◇ **n = 15**: Sample size per group and Total sample size = 30 (15 in each group)
- ◇ **delta = 1**: True difference between population means $\delta = \mu_1 - \mu_2$
- ◇ **sd = 1**: Assumed standard deviation in each group
- ◇ **alternative = "two.sided"**

Testing:

$$H_0: \mu_1 = \mu_2$$

$$H_1: \mu_1 \neq \mu_2$$

- ◇ **Power=0.7529** : There is a **75.3% probability** of correctly rejecting H_0 if the true difference is 1.

👉 Our experiment with 15 subjects per group, Effect size = 1 at significance level 0.05 with 75.3% power

Meaning the study had a reasonably good but slightly insufficient ability to detect the true difference.



A Priori Power Analysis (Calculating Required Sample Size for a t-test in R)

◇ Part 1: Calculate number of replication for `t.test` in R.

A priori power calculation: Extract information from literature or pilot study and calculate n for a given power.

R Code:

➤ `power.t.test(power = 0.8, delta = 1, sig.level = 0.05, alternative = "two.sided")`

Output Interpretation

Two-sample t test power calculation

n = 16.71477

delta = 1

sd = 1

sig.level = 0.05

power = 0.8

alternative = two.sided

NOTE: n is number in *each* group



Each Parameter Means:

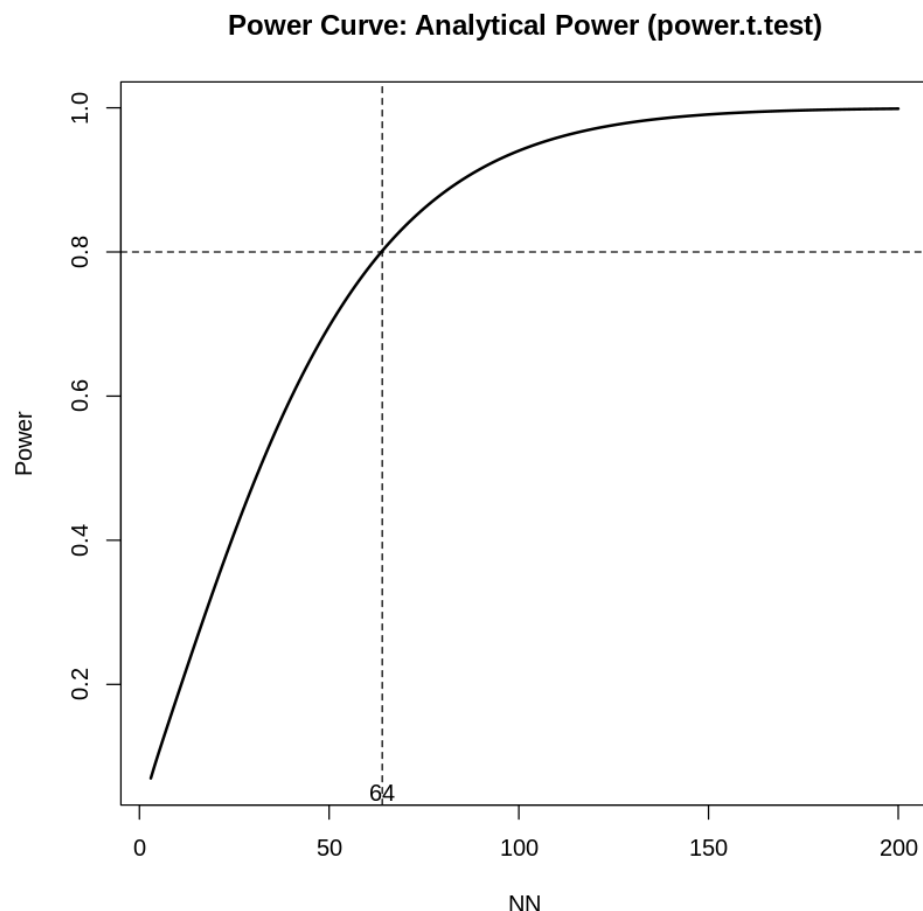
- ◇ **$n = 16.71477$** : Since we cannot have fractional subjects:
 $n \approx 17$, So we need: 17 subjects per group and Total sample size = 34
- ◇ **$\delta = 1$** : Expected true difference between the two group means.
Usually obtained from:
 - Literature
 - Similar studies
 - Pilot experiment
- ◇ **$sd = 1$** : Estimated standard deviation.
Again obtained from:
 - Previous studies
 - Pilot data
- ◇ **$\text{sig.level} = 0.05$** : Type I error: $\alpha = 0.05$
- ◇ **Power = 0.8** : Power = 80% This means: $\beta = 1 - 0.8 = 0.2$. So we accept a 20% probability of Type II error.

 To detect a true difference of 1 (with SD = 1) at significance level 0.05 with 80% power, we need at least: 17 observation per group.



Power Curve using `power.t.test()`

To compute power for a range of sample sizes and then plot Power vs Sample Size.



Computes power for $n = 1$ to 200 (per group)

$$\text{Power} = P(\text{Reject } H_0 \mid H_1 \text{ TRUE}) = 1 - \beta$$

Draws Power Curve

- X-axis → Sample size per group
- Y-axis → Power
- Curve increases and approaches 1

Interpretation of the Graph

- Small $n \rightarrow$ low power
- Around $n \approx 64 \rightarrow \text{Power} \approx 0.8$
- Large $n \rightarrow$ power approaches 1

The dashed horizontal and vertical lines in your figure correspond to:

Power=0.8 and $n \approx 64$



Power Calculation using Simulation Method

This is very useful when:

- Analytical formulas are complicated
- Assumptions are unclear
- You want to demonstrate power conceptually to students

To estimate the **power of a two-sample t-test by simulating data.**

In this study, 15 males and 15 females were sampled from normally distributed populations with mean lengths of 18 and 17 cm, respectively, and a standard deviation of 2 cm for both groups

Given:

- Males: mean = 18 cm
- Females: mean = 17 cm
- Standard deviation = 2 cm
- Sample size = 15 per group
- True difference = 1 cm

We simulate data and check how often the t-test detects this difference.



Step 1: Simulating the Data

- `set.seed(658)`
- `Males <- round(rnorm(n = 15, mean = 18, sd = 2),2)`
- `Females <- round(rnorm(n = 15, mean = 17, sd = 2),2)`

Generates random values from a normal distribution.

So we assume:

$$X_{\text{Male}} \sim N(18, 2^2)$$

$$X_{\text{Female}} \sim N(17, 2^2)$$



Step 2: Perform the t-test

Part 1: One t-test on Simulated Samples

we can perform a two-sided t-test with equal variance.

We generated:

- Males $\sim N(18, 2^2)$
- Females $\sim N(17, 2^2)$
- Sample size = 15 per group

➤ `t.test(Males, Females, var.equal = TRUE)`

Output:

t = 1.6207

df = 28

p-value = 0.1163

95% CI: -0.356 to 3.052

mean of x = 18.43

mean of y = 17.09

Sample Means

Observed difference: $18.43 - 17.09 = 1.34$

True population difference = 1

Sample difference = 1.34

p-value = 0.1163

Since: $p = 0.1163 > 0.05$

We **fail to reject** H_0 .

Even though a true difference exists, this particular sample did not detect it.

☞ This is a **Type II error** in this specific simulated experiment.

Confidence Interval: (-0.356, 3.052)

Since 0 lies inside the interval → not significant.



◇ Part 2: 10,000 Simulated t-tests (n = 20)

Calculate the p-values of 10000 t-tests for n = 20, effect size 18-17 and SD 2.

Now we simulate the experiment 10,000 times.

```
➤ pval <- numeric(10000)

➤ for(i in 1:10000){
  A <- rnorm(20, 18, 2)
  B <- rnorm(20, 17, 2)
  pval[i] <- t.test(A,B)$p.value
}
```

pval contains 10000 p-values. Now we calculate how many p-values are smaller than 0.05 and divide this number by 10000, which is the power.

```
➤ sum(pval < 0.05)/10000
```

Output: 0.3419

Interpretation : Power $\approx$ 0.3419 (34.2%)

Meaning: $P(\text{Reject } H_0 \mid \text{True difference} = 1) = 34\%$

So $\beta = 1 - 0.3419 = 0.6581$

There is about **66% chance of missing the true effect.**



◇ Part 3: 10,000 Simulated t-tests (n = 40)

Calculate the p-values of 10000 t-tests for n = 40, effect size 18-17 and SD 2.

```
➤ pval <- numeric(10000)

➤ for(i in 1:10000){
  A <- rnorm(40, 18, 2)
  B <- rnorm(40, 17, 2)
  pval[i] <- t.test(A,B)$p.value
}
```

Calculating the power using pval as in previous slide.

```
sum(pval < 0.05)/10000
```

Output: 0.6021

Interpretation: Power $\approx$ 0.60 (60%)

Meaning: $P(\text{Reject } H_0 \mid \text{True difference} = 1) = 60\%$

So $\beta = 1 - 0.6021 = 0.3979$

Type II error drops from 66% $\rightarrow$ 40%.



Why Did Power Increase?

Because increasing sample size:

- Reduces standard error
- Makes distributions narrower
- Reduces overlap
- Makes it easier to detect difference

Mathematically:

$$SE = \frac{\sigma}{\sqrt{n}}$$

As $n \uparrow$, $SE \downarrow$

Sample Size (per group)	Power
15	~ 0.11 (in single example)
20	0.34 (34%)
40	0.60 (60%)

Still not at recommended 80%.

To reach 80%, you would need even larger n.



Power Curve Using Simulation Method (Nested Loops)

- First loop for 198 different sample sizes.
- Second loop, nested in first loop, to create 1000 t-tests.

This uses:

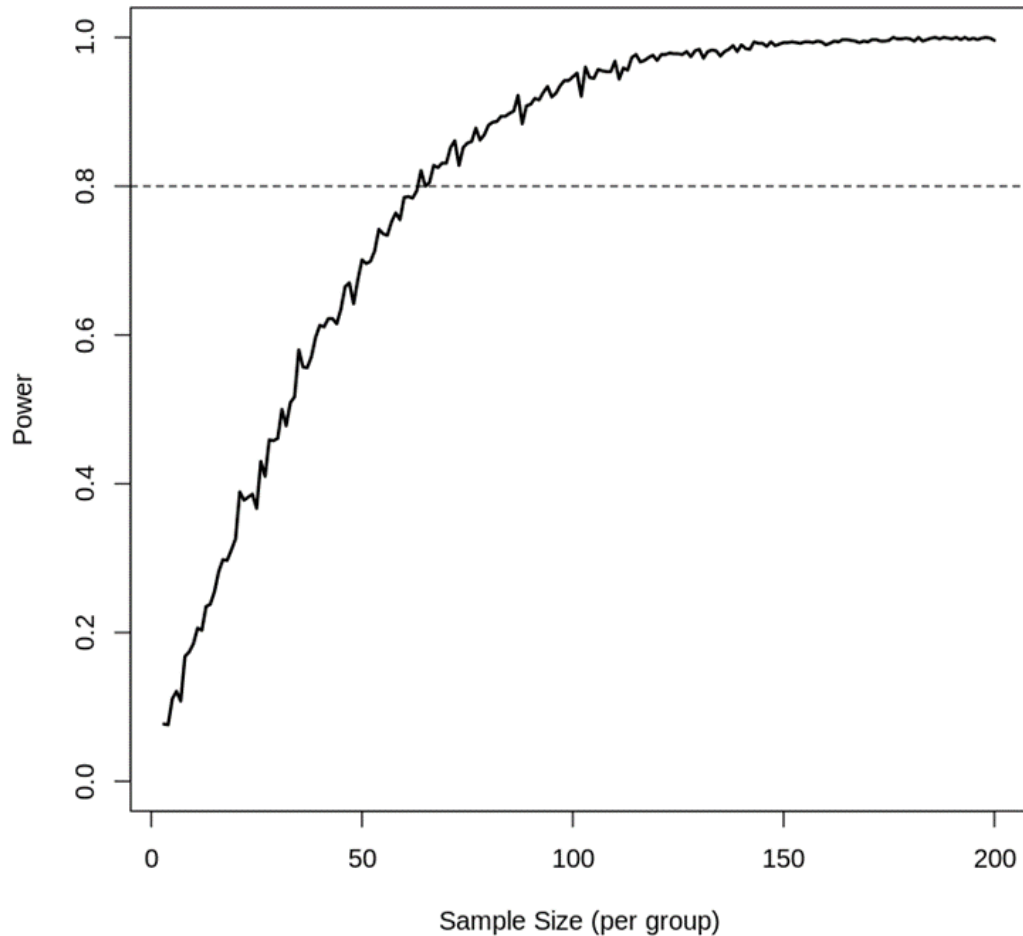
- 198 different sample sizes (3 to 200)
- 1000 simulated t-tests for each sample size
- Effect size: $18 - 17 = 1$
- SD = 2
- Two-sided test
- $\alpha = 0.05$

```
Power2 <- numeric(198)      # Placeholder for power values
pval <- numeric(1000)       # Placeholder for p-values from 1000 simulations
NN <- 3:200
# Sample sizes
for(j in 1:198){            # First loop for 198 sample sizes
  for(i in 1:1000){          # Second loop for 1000 tests.
    A<-rnorm(NN[j], 18, 2)    # Sample A
    B<-rnorm(NN[j], 17, 2)    # Sample B
    pval[i]<-t.test(A,B)$p.value # Actual t-test
  }                          # End of second loop of 1000 tests
  Power2[j] <- sum(pval< 0.05)/1000 # Computes power over 1000 tests
}                             # End of first loop of 198 sample sizes.
```



Power Curve Using Simulation (Nested Loops)

Power Curve (Simulation)



The Curve Shows

- Small $n \rightarrow$ low power
- As n increases $\rightarrow$ power increases
- Curve approaches 1
- Around $n \approx 64$ per group $\rightarrow$ power ≈ 0.8 (approximately)



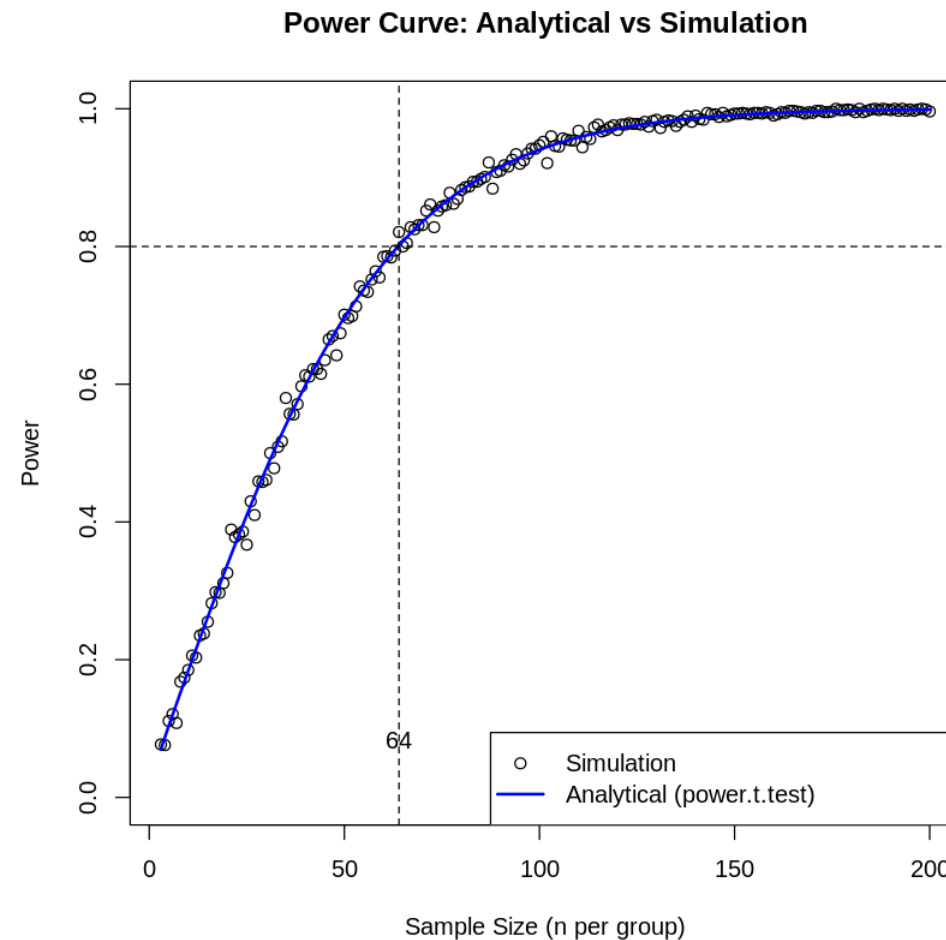
Combined Power Curve (Analytical + Simulation)

To generate the **combined power curve** using:

1. **Analytical method** → `power.t.test()` (theoretical power)
2. **Simulation method** → Nested loops (1000 t-tests per sample size) i.e. Simulation with nested loops (empirical power)

PARAMETERS:

- Two-sample t-test
- Mean1 = 18
- Mean2 = 17
- sd = 2
- $\alpha = 0.05$
- Two-sided test
- Sample sizes: 3 to 200
- 1000 simulations per sample size





The graph Shows

- Black circles → Simulated power (1000 t-tests per sample size)
- Blue line → Exact theoretical power from `power.t.test()`
- Horizontal dashed line → 80% power
- Vertical dashed line → Required sample size per group

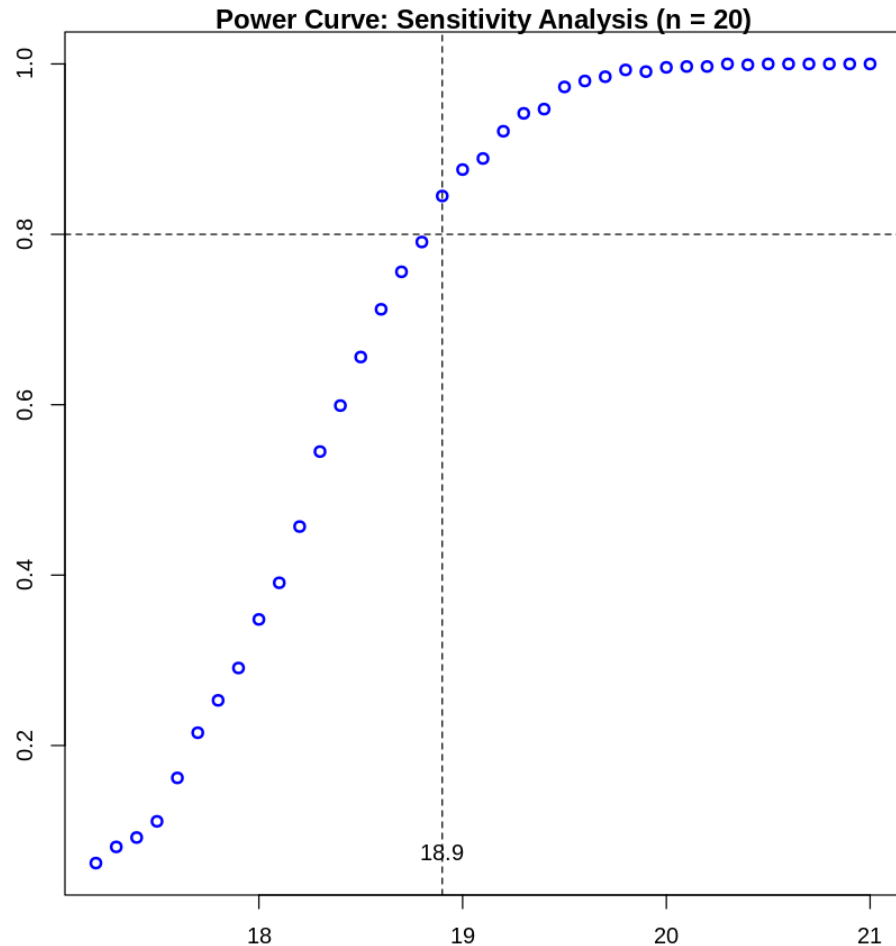
Interpretation

- The two curves should almost overlap (as in your image).
- Small differences occur due to Monte Carlo simulation error.
- The required n per group for 80% power is around **64** (as seen in your figure).



Power Curve for Effect Size (Simulation)

The **Power Curve for Effect Size (Sensitivity Analysis)** with fixed sample size $n=20$ per group



- X-axis → Effect Size (mean difference)
- Y-axis → Power
- Dots → Simulated power values
- Horizontal dashed line → 80% power
- Vertical dashed line → Effect size needed for 80% power (~18.9 in your image)

Interpretation

- With $n = 20$ per group and $sd = 2$:
- Small effect sizes → low power
- Around mean $\approx 18.8 \rightarrow$ power ≈ 0.8
- Large effect sizes → power $\rightarrow 1$

This is called **Sensitivity Analysis** because we fix sample size and examine what effect size we can detect.

Replications and Experimental Units

Which observations count as an independent measurements; the experimental unit.

- **Independence:** The experimental unit is not influenced by treatments to other units.
- **Treatment Application:** The unit receives the treatment or condition this is being tested.
- **Observation:** Data are collected from the experimental unit after the treatment is applied
- **Experimental Unit:** The experimental unit is the smallest division of material in an experiment where any two units can receive different treatments.
- **Sample Unit:** All sample units (individual measurements) within an experimental unit receive the same treatment, and thus share the defining characteristics of that experimental unit.

For example, six sample units within a “confinement” (such as a location or subject) receive the same treatment, the confinement is considered the experimental unit. “Confinement” could refer to a specific locations, like plots in a field, or to a subject, such as a patient or a plant

Question 1:

Every rat was injected with a Placebo or vitamin C.

How many replicates do we have, and what is the experimental unit?

Note

All measurements were done independently; every animal (sample unit) gets its own treatment (injection) with all the artefacts of the act of injection. The animal is the experimental unit. There are 6 independent measurements in this experiment 3 for A and three for B that did not influenced each other

Answer

- **Experimental unit:** Each rat
- **Number of replicates:** 3 per treatment (6 total independent observations)

Since each rat was treated and measured independently, there is no pseudo replication.

Question 2:

In each box, two rats were given a diet of either water or water supplemented with vitamin C.

How many replicates are there, and what is the experimental unit?

The two rats in a box are fed the same diet particular to that box; small variation in vitamin C compared to the other boxes, the other rat or temperature due to the location in the room.

As a result the boxes are the experimental unit and the number of independent measurements are therefore 2 for water and 2 for vitamin with water in total 4 independent measurements.

Answer

- **Experimental unit:** Box
- **Replicates:** 2 per treatment
- **Total independent measurements:** 4

Question 3:

It is hypothesized that human aggression is linked to resting adrenaline levels in the blood. To investigate this, a researcher uses two rat species with known differences in their resting adrenaline concentrations to determine if these variations influence aggression. How many experimental units are involved in this study?

Indeed the two strain might differ in basic adrenaline levels, we say that the adrenaline treatment is nested in strain. Strain are the experimental unit.

Answer

- **Experimental unit:** Strain
- **Number of experimental units:** 2
- Adrenaline is **nested within strain**, not independently manipulated.

Question 4:

To investigate the effect of vitamin C on liver fat content, rats A to C were given a standard vitamin c diet and rats D-F a high vitamin C diet. At the end of the treatment the rats were sacrificed and their liver chopped in 6 pieces for fat content analysis. How many replicates do we have and what is the experimental unit?

Every piece of liver from one rat liver is genetically and historically related to that rat. If the rat had a liver disease than all 6 pieces could be affected. The experimental unit here is the subject rat; liver pieces are nested in rat

Answer

- **Experimental unit:** Rat
- **Replicates:** 3 per treatment
- **Total independent measurements:** 6
- **Liver pieces are nested within rat** (not independent)

Question 5:

From 5 different cultivars we randomly select three leaves and measure the ration between dry weight and water content. How many experiment units do we have?

Answer

- **Experimental unit:** Cultivar
- **Number of experimental units:** 5
- **Leaves are nested within cultivar**

- <https://www.tidyverse.org/>
- <https://www.tidymodels.org/>
- <https://cran.r-project.org/web/packages/broom/vignettes/broom.html>

R packages for data science

- **TIDYVERSE**

➤ `install.packages("tidyverse")`

- **TIDYMODELS**

➤ `install.packages("tidymodels")`

R-NOTEBOOK

<https://github.com/Mrinalini0107/Experimental-Design-Using-R/tree/main>



Experimental Design

Thank You ...!